

LumenCore Capability Note for 693JJ326R000012

FHWA Transportation Systems Management and Operations Data Initiative

Entity	Robert Ashworth / LumenCore
UEI	SQY2XW71ZM51
Primary NAICS	541511 - Custom Computer Programming Services
Response posture	Capability/intent note; not a final proposal or price submission
Public proof gateway	https://lumen-core.ai/proof_to_pilot.html

Concept Fit

LumenCore is building a proof-to-pilot validation layer for AI-enabled infrastructure workflows. The platform focuses on source-quality gates, baseline-vs-candidate replay, evidence manifests, and reviewer-readable proof cards.

The FHWA TSMO Data Initiative appears aligned with identifying data barriers to AI adoption, prioritizing use cases, and prototyping evidence-bounded algorithms or models for transportation operations decision support.

Potential Work Areas

- Data readiness scoring for TSMO feeds: freshness, missingness, schema drift, conflict rate, lineage, and reproducibility.
- Use-case prioritization method that ranks AI opportunities by data readiness, operational value, validation burden, and safety constraints.
- Baseline-compared AI incident, congestion, anomaly, or operations-review workflows using pre-registered metrics.
- Machine-readable evidence manifests and human-readable proof cards for technical reviewers.

Claim Boundary

LumenCore can contribute a focused validation/prototype layer and evidence discipline. LumenCore is not claiming existing FHWA field validation, operational transportation deployment, safety certification, realized savings, FedRAMP authorization, or award status.

Requested Next Step

Please confirm the official submission instructions, required attachments/volumes, any amendment package, and whether a focused small-business validation/prototype role is responsive to the Phase I technical capability path.